4. Summary tables of information

Key:

- This vaccination is safe to be given. It may be routinely recommended or given if clinically indicated
- This vaccination is contraindicated in pregnancy or the safety of the vaccination in these populations has not been determined
- This vaccination is recommended if international travel is required

Table 1. Vaccination recommendations- Australia

Disease/Vaccination	Planning pregnancy	Trimester 1	Trimester 2	Trimester 3	Postpartum (including breastfeeding)	Infants		
						2 months	4 months	6 months
COVID-19	Recommended for people 5 years and older.[i]	Strongly recommended[i] (mRNA type only). Safe to give at any stage of pregnancy and postpartum for women of any age. Women who are pregnant should stay up-to-date with vaccinations.[ii]				Not indicated.[i]	Not indicated.[i]	Recommended for infants aged 6 months with severe immunocompromise, disability and those at greatest risk of severe outcomes from COVID-19 only.
Hepatitis A	Not routinely recommended, unless non-immune and at increased risk, including travel to area of moderate to high endemicity or increased risk of exposure and/or expected severe outcomes (lifestyle factors, pre-existing liver disease).[iii]					Not indicated < 12 months of age.		

[i] Please note, advice regarding the number and type of recommended additional doses of approved COVID-19 vaccines is subject to change. Please refer to COVID-19 vaccines | Australian Government Department of Health and Aged Care and the COVID-19 vaccination in pregnant and breastfeeding women and those planning pregnancy - RANZCOG advisory bulletin (information current at publication).

[ii] See ATAGI statement on defining 'up-to-date' status for COVID-19 vaccination, 10 February 2022 (information current at publication)- https://www.health.gov.au/news/atagi-statement-on-defining-up-to-date-status-for-covid-19-vaccination.

[iii] Australian Technical Advisory Group on Immunisation (ATAGI). Australian Immunisation Handbook- Vaccines that are not routinely recommended in pregnancy: inactivated viral vaccines. Canberra, 2018. Table. Vaccines that are not routinely recommended in pregnancy: inactivated viral vaccines | The Australian Immunisation Handbook (health.gov.au)

Table 1. Vaccination recommendations- Australia

Disease/Vaccination	Planning pregnancy	Trimester 1	Trimester 2	Trimester 3	Postpartum (including breastfeeding)	Infants		
						2 months	4 months	6 months
Hepatitis B	Women planning pregnancy should be up to date with this vaccination.	✓ Not routinely recommended. Pregnant women **can receive if non-immune and at increased risk for hepatitis B**.				✓ Plus a dose at birth (dose 0). Recommended as early as 6 weeks of age (usually given with DTPa-hepB-IPV-Hib for additional doses).[iv]	✓	✓
Haemophilus influenzae type b (Hib-PRP) disease	Only recommended if high risk and clinically indicated.	✓ Not routinely recommended. Pregnant women **can receive Hib vaccine if at increased risk.**[viii]				✓ As early as 6 weeks of age.	✓ Doses 1-3 given as DTPa-hepB-IPV-Hib combination vaccine.[xiii]	✓
Human papillomavirus (HPV)	Not routinely recommended for adults >19 years of age (however risk profile should be considered for eligibility).	Not recommended. Women who become pregnant after starting HPV vaccination course should stop and receive the remaining doses after pregnancy.			✓ Breastfeeding women can receive HPV vaccines **if required**.	Not indicated < 9 years of age.		

[iv] Australian Technical Advisory Group on Immunisation (ATAGI). Australian Immunisation Handbook- Hepatitis B: Recommendations- Infants, children and adolescents. Canberra, 2021. https://immunisationhandbook.health.gov.au/contents/vaccine-preventable-diseases/hepatitis-b#infants-children-and-adolescents

Table 1. Vaccination recommendations- Australia

Disease/Vaccination	Planning pregnancy	Trimester 1	Trimester 2	Trimester 3	Postpartum (including breastfeeding)	Infants		
						2 months	4 months	6 months
Influenza	✓ Strongly recommended.[viii]	✓ Recommended for all pregnant women at any trimester **in *each* pregnancy**, especially if pregnant during influenza season (autumn/winter).[viii]			✓ Recommended and safe to receive during breastfeeding.[viii]	Not indicated- maternal immunisation may also provide infant with protective antibodies.		✓ Infants can receive first Influenza vaccine from 6 months of age (2 doses- 4 weeks apart within 12 months of age).
Japanese encephalitis (JE)	Not routinely recommended. Inactivated vaccine is considered safe to use in women planning pregnancy.[v]	✓ Not routinely recommended for pregnancy or breastfeeding women. The Imojev (live attenuated) vaccine is contraindicated in pregnancy and breastfeeding. However, **pregnant women at high risk of acquiring JE can receive inactivated vaccines.**[vi] ✈ Also recommended for pregnant travellers spending 1 month or more in endemic areas (Asia, PNG) during JE transmission season.[v]				✓ ✈ If indicated. Inactivated JE vaccines can be administered to infants from ≥2 months of age.		
Measles and mumps	✗ Avoid pregnancy within 28 days of receiving vaccination.	✗ Contraindicated during pregnancy. Hypothetical risk only- see Additional Practical Advice. All pregnant women are recommended to be tested for immunity to rubella and varicella.[vii]			✓ Susceptible women should receive the rubella vaccine as soon as possible after delivery and have serological status	Not indicated < 12 months of age.		

[v] Australian Technical Advisory Group on Immunisation (ATAGI). Australian Immunisation Handbook- Japanese-encephalitis- Contraindications and precautions: Women who are pregnant or breastfeeding. Canberra, 2021. Japanese encephalitis | The Australian Immunisation Handbook (health.gov.au)

[vi] NB: No specific data are available about using JEspect (inactivated vaccine) in breastfeeding women. https://immunisationhandbook.health.gov.au/contents/vaccine-preventable-diseases/japanese-encephalitis#women-who-are-pregnant-or-breastfeeding

[vii] Australian Technical Advisory Group on Immunisation (ATAGI). Australian Immunisation Handbook- Vaccines that are contraindicated in pregnancy: live attenuated vaccines. Canberra, 2018. Table. Vaccines that are contraindicated in pregnancy: live attenuated vaccines | The Australian Immunisation Handbook (health.gov.au)

Table 1. Vaccination recommendations- Australia

Disease/Vaccination	Planning pregnancy	Trimester 1	Trimester 2	Trimester 3	Postpartum (including breastfeeding)	Infants		
						2 months	4 months	6 months
					checked after vaccination.[vii]			
Meningococcal disease Meningococcal B	Only recommended if at increased risk.	Not routinely recommended. Pregnant women can receive if clinically indicated.[viii]				✓ As early as 6 weeks of age.[ix]	✓ 8 weeks required between 1st and 2nd doses.	3rd dose not required until 12 months of age.
Monkeypox (mpox)[x]	Recommended for anyone categorised by public health authorities as high risk mpox contact <14 days.[x]	Not routinely recommended. The JYNNEOS® vaccine (replication-deficient) is considered safe to use in women who are pregnant and a high risk mpox contact <14 days. Risk-benefit analysis recommended.[x]				✗ Not recommended due to limited studies available. Both vaccines are contraindicated for infants < 12 months of age.[x]		
Pertussis (whooping cough) dTpa combined vaccine-diphtheria, tetanus[xi], and acellular pertussis-containing vaccine	No advice identified specific to planning pregnancy.	✓ Recommended as a single dose **in *each* pregnancy** between mid-2nd trimester and early 3rd trimester (preferably between 20-32 weeks).			✓ Vaccination is recommended **as soon as possible after birth** if not vaccinated during pregnancy and >10 years since last dose.[xii]	✓ As early as six weeks of age.	✓ One dose at 4 months, another dose at 6 months. Given as DTPa-hepB-IPV-Hib combination vaccine- diphtheria-tetanus-acellular pertussis,	

[viii] Australian Technical Advisory Group on Immunisation (ATAGI). Australian Immunisation Handbook- Vaccination for women who are planning pregnancy, pregnant or breastfeeding- Table: Vaccines that are routinely recommended in pregnancy. Canberra, 2021. https://immunisationhandbook.health.gov.au/contents/vaccination-for-special-risk-groups/vaccination-for-women-who-are-planning-pregnancy-pregnant-or-breastfeeding#recommended-vaccines-for-pregnant-women

[ix] Please note, the Meningococcal B vaccine is **not free** for non-Indigenous children aged 2 months, 4 months, and 6 months under the National Immunisation Program, however the vaccine is still strongly recommended.

[x] Please note the ATAGI Clinical Guidance on vaccination against monkeypox did not undergo a quality assessment using the AGREE II Framework. This resource is provided as it is currently absent from the Australian Immunisation Handbook, likely owing to the recency of the 2022 multi-country outbreak. See Updated ATAGI Clinical Guidance on Vaccination against Monkeypox (MPOX) Version 4 (current as at publication of statement)- https://www.health.gov.au/sites/default/files/2022-12/atagi-clinical-guidance-on-vaccination-against-monkeypox.pdf

[xi] There is no minimum interval between a dose of dT (tetanus and diphtheria; for wound management) and dTpa when dTpa is being given for pertussis protection during pregnancy.

[xii] See Australian Immunisation Handbook- Tetanus for further information on revaccination prior to 10 years since last dose- https://immunisationhandbook.health.gov.au/contents/vaccine-preventable-diseases/tetanus

Table 1. Vaccination recommendations- Australia

Disease/Vaccination	Planning pregnancy	Trimester 1	Trimester 2	Trimester 3	Postpartum (including breastfeeding)	Infants		
						2 months	4 months	6 months
							hepatitis B, inactivated poliovirus, *Haemophilus influenzae* type b).	
Pneumococcal disease	✓ Women of child-bearing age with a risk condition for pneumococcal disease can receive the vaccine before a planned pregnancy.	Not routinely recommended. Pregnant women with a risk for invasive pneumococcal disease (IPD), **can receive the vaccine if required**.			✓ Women of child-bearing age with a risk for pneumococcal disease can receive the vaccine **as soon as practicable** after birth.[viii]	✓ As early as 6 weeks of age.	✓	3rd dose not required until 12 months of age.
Poliomyelitis	⚠ Not routinely recommended. If travelling to **endemic country or at high risk of poliovirus exposure**, pregnant women can receive IPV vaccine.[iii]					✓ Given as DTPa-hepB -IPV-Hib combination vaccine for Doses 1-3.[xiii]	✓	✓
Q Fever	Only if indicated. Safe use in women planning pregnancy not established.	✗ Not routinely recommended. Safe use of Q fever vaccine in pregnancy and while breastfeeding has not been established.				✗ Not recommended for off-label use in people < 15 years of age.		
Rabies	Not routinely recommended but **registered for use in any age**.	✓ Not routinely recommended but safe to be administered during pregnancy and while breastfeeding if required, i.e., as part of post-exposure prophylaxis.[iii]				Not routinely recommended but registered for use in any age. Consider based on individual risk-benefit balance.		

[xiii] Australian Technical Advisory Group on Immunisation (ATAGI). Australian Immunisation Handbook- Poliomyelitis: Infants and children, Australian Government Department of Health and Aged Care, Canberra, 2021. https://immunisationhandbook.health.gov.au/contents/vaccine-preventable-diseases/poliomyelitis#infants-and-children

Table 1. Vaccination recommendations- Australia

Disease/Vaccination	Planning pregnancy	Trimester 1	Trimester 2	Trimester 3	Postpartum (including breastfeeding)	Infants		
						2 months	4 months	6 months
Respiratory syncytial virus (RSV)	No recommendation			✓ A single dose of Abrysvo is recommended for pregnant women at 28–36 weeks gestation to protect the infant.	No recommendation	✓ Nirsevimab (long-acting RSV-specific monoclonal antibody) is recommended for infants who were born: • to women who did not receive RSV vaccine during pregnancy • within 2 weeks of the mother receiving RSV vaccine during pregnancy • Higher risk infants after assessment by treating doctor to confirm potential clinical benefit.[xiv] Timing of administration of Nirsevimab should ensure duration and level of protection are maximised over the peak months of a child's 1st RSV season.[xv] ✗ Nirsevimab is **not** recommended for infants during first 6 months of life if: • infant's mother received RSV vaccine at an appropriate time during pregnancy, and • infant does not have a risk condition for severe RSV disease		
Rubella (given as MMR)	✓ Immunisation providers **should make every effort to identify and immunise non-pregnant**	✗ MMR- containing vaccines are contraindicated in pregnant women.[xx]			✓ Breastfeeding women can receive MMR vaccines.	Not indicated < 12 months of age.		

[xiv] Higher risk includes- infants with risk conditions for severe RSV, regardless of maternal vaccination (see List); infants born to mothers with severe immunosuppression where immune response to maternal RSV vaccination impaired (see List); infants whose mothers have received RSV vaccination in pregnancy but have subsequently undergone a treatment such as cardiopulmonary bypass or extracorporeal membrane oxygenation, that has led to loss of maternal antibodies.

[xv] Definition of RSV season varies- typically April to September in temperate regions of Australia. Local advice should be sought. See sub-section Timing of RSV-specific monoclonal antibodies in infants.

Table 1. Vaccination recommendations- Australia

Disease/Vaccination	Planning pregnancy	Trimester 1	Trimester 2	Trimester 3	Postpartum (including breastfeeding)	Infants		
						2 months	4 months	6 months
	seronegative women of child-bearing age. Pregnancy should be avoided for 28 days after vaccination.[xvi]				Postpartum vaccination of non-immune women does not need to be delayed due to breastfeeding.[xx]			
Rotavirus	Childhood immunisation only.					✓ Rotarix.	✓ Rotarix.	✓ Only required if vaccination course of RotaTeq is given.
Typhoid fever	✈ Not routinely recommended, unless travelling to endemic regions.					Not indicated < 2 years of age.		
Varicella (chickenpox)	✓ Immunisation providers should make **every effort to identify and immunise** non-pregnant seronegative women of childbearing age. Avoid pregnancy for 28 days after vaccination.[xvii]	✗ Varicella- containing vaccines are contraindicated in pregnant women.[xvii]			✓ Non-immune breastfeeding women can receive varicella vaccine. No effects on breastfed infants have been reported.[xvii]	Not recommended for infants <2 years of age.		

[xvi] Australian Technical Advisory Group on Immunisation (ATAGI). Australian Immunisation Handbook- Rubella- Pregnant women, Australian Government Department of Health and Aged Care, Canberra, 2022. https://immunisationhandbook.health.gov.au/contents/vaccine-preventable-diseases/rubella#pregnant-women

[xvii] Australian Technical Advisory Group on Immunisation (ATAGI). Australian Immunisation Handbook- Varicella (chickenpox)- Women who are pregnant or breastfeeding, Australian Government Department of Health and Aged Care, Canberra, 2022. https://immunisationhandbook.health.gov.au/contents/vaccine-preventable-diseases/varicella-chickenpox#women-who-are-pregnant-or-breastfeeding

Table 1. Vaccination recommendations- Australia

Disease/Vaccination	Planning pregnancy	Trimester 1	Trimester 2	Trimester 3	Postpartum (including breastfeeding)	Infants		
						2 months	4 months	6 months
Yellow fever	Not routinely recommended- only if travelling to area with a risk of yellow fever virus transmission.	Not recommended. Pregnant women should be advised to avoid going to rural areas where yellow fever transmission is a risk. However, **if travel is unavoidable**, pregnant women **should** receive the vaccine.[xviii] [xix]			Should be avoided during breastfeeding (< 9 months) unless mother has high risk of acquiring yellow fever or cannot avoid/postpone travel.[xix]	Contraindicated in infants <9 months of age.[xx]		

[xviii] Please note, women who are pregnant at the time of receiving their yellow fever vaccine may be required to receive a booster dose if they do not respond optimally to a single dose. See- https://immunisationhandbook.health.gov.au/contents/vaccine-preventable-diseases/yellow-fever#travellers

[xix] Australian Technical Advisory Group on Immunisation (ATAGI). Australian Immunisation Handbook- Yellow fever: Women who are pregnant or breastfeeding, Australian Government Department of Health and Aged Care, Canberra, 2022, https://immunisationhandbook.health.gov.au/contents/vaccine-preventable-diseases/yellow-fever#women-who-are-pregnant-or-breastfeeding

[xx] Australian Technical Advisory Group on Immunisation (ATAGI). Australian Immunisation Handbook- Yellow fever- Infants, Australian Government Department of Health and Aged Care, Canberra, 2022. https://immunisationhandbook.health.gov.au/contents/vaccine-preventable-diseases/yellow-fever#infants

Table 2. Vaccination recommendations- Aotearoa New Zealand

Disease/Vaccination	Planning pregnancy	Trimester 1	Trimester 2	Trimester 3	Postpartum	Infants		
						6 weeks	3 months	5 months
COVID-19	✓ Anyone who is pregnant, or planning pregnancy is encouraged to be routinely vaccinated with mRNA-CV at any stage of pregnancy. [xxi]				✓ There are no safety concerns about giving mRNA-CV to those lactating.[xxi]	Not available for individuals < 5 years of age.		
Haemophilus influenzae type b (Hib) disease	Only if high risk. [xxii]	Not routinely recommended for pregnant or breastfeeding women.[xxii]				✓	✓	✓
Hepatitis A	Only if high risk. [xxiii]	✕ ✈ Safety of HepA during pregnancy and while breastfeeding has not yet been determined,[xxiii] however as an inactivated vaccine, there is not expected to be any risk to the developing fetus and infant.[xxiii] As a precaution, the HepA **vaccine should only be used during pregnancy when clearly needed** (i.e., travel to a high-risk country).				Not routinely recommended. Only if immunosuppressed or during community outbreak.[xxiv]		

[xxi] Manatu Hauora (Ministry of Health) Aotearoa New Zealand. Immunisation Handbook- COVID-19. Section 5.5.3 Breastfeeding and 5.5.4- Pregnancy: Online, 2020. 5. Coronavirus disease (COVID-19) | Ministry of Health NZ

[xxii] Manatu Hauora (Ministry of Health) Aotearoa New Zealand. Immunisation Handbook- Hib Disease. Section 7.5- Recommended immunisation schedule and Section 7.5.5- Pregnancy and breastfeeding: Online, 2020. 7. Haemophilus influenzae type b (Hib) disease – Immunisation Handbook 2020 | Ministry of Health NZ

[xxiii] Manatu Hauora (Ministry of Health) Aotearoa New Zealand. Immunisation Handbook- Hepatitis A. Section 8.5.1- Recommendations and Section 8.5.3- Pregnancy and breastfeeding: Online, 2020. 8. Hepatitis A – Immunisation Handbook 2020 | Ministry of Health NZ

[xxiv] Manatu Hauora (Ministry of Health) Aotearoa New Zealand. Immunisation Handbook- Hepatitis A. Section 8.5.1 Recommendations- Routine immunisation for children and Section 8.5.2 Immunisation Schedule. Online, 2020. 8. Hepatitis A – Immunisation Handbook 2020 | Ministry of Health NZ

Table 2. Vaccination recommendations- Aotearoa New Zealand

Disease/Vaccination	Planning pregnancy	Trimester 1	Trimester 2	Trimester 3	Postpartum	Infants		
						6 weeks	3 months	5 months
Hepatitis B	Women planning pregnancy should be up to date with this vaccination.	✓ Pregnant women should be screened and can receive during pregnancy and while breastfeeding **if non-immune and at increased risk** for acute HBV infection.[xxv]				✓ Plus a dose at birth for infants born to carrier mothers.[xxv] Recommended at 6 weeks of age.	✓	✓
Human papillomavirus (HPV)	Women <26yo are eligible for the HPV vaccine. Enquiring about possibility of woman being pregnant is not necessary before vaccination (no adverse effects shown on pregnancy outcomes).[xxvi]	Not recommended. Women who become pregnant after starting HPV vaccination course **should be delayed until after pregnancy.**[xxvi]			✓ Breastfeeding women can receive HPV vaccines.	Not indicated < 9 years of age.		

[xxv] Manatu Hauora (Ministry of Health) Aotearoa New Zealand. Immunisation Handbook- Hepatitis B. Section 9.5 Recommended immunisation schedule and Section 9.5.5- Pregnancy and breastfeeding. Online, 2020. 9. Hepatitis B – Immunisation Handbook 2020 | Ministry of Health NZ

[xxvi] Manatu Hauora (Ministry of Health) Aotearoa New Zealand. Immunisation Handbook- HPV. Section 10.5- Recommended immunisation schedule and Section 10.5.3 Pregnancy and breastfeeding. 10. Human papillomavirus – Immunisation Handbook 2020 | Ministry of Health NZ

Table 2. Vaccination recommendations- Aotearoa New Zealand

Disease/Vaccination	Planning pregnancy	Trimester 1	Trimester 2	Trimester 3	Postpartum	Infants		
						6 weeks	3 months	5 months
Influenza	✓ Recommended for all adults.	✓ Recommended for all pregnant women at any trimester, especially if pregnant during influenza season (autumn/winter, typically May to September).[xxvii]			✓ Recommended and safe to receive during breastfeeding.[xxvii]	Not indicated. Maternal-fetal passive antibody transfer provides infant with some protection.[xxvii]		Only indicated from 6 months of age if infant has chronic illness or history of significant respiratory disease.[xxvii]
Measles and mumps	✗ Avoid pregnancy for four weeks after MMR vaccination.[xxviii]	✗ Contraindicated during pregnancy[xxix] Follow-up studies of women who inadvertently received MMR during pregnancy show **no evidence MMR is teratogenic** or harmful to the mother, her fetus or her newborn. All pregnant women are recommended to be tested for immunity to rubella.			✓ Can be given safely after birth (if required) and to breastfeeding women.[xxix]	Not indicated < 12 months of age.		
Meningococcal disease Meningococcal B	Not routinely recommended. Women can receive prior to or during pregnancy if indicated. No adverse effects among pregnant women vaccinated reported.[xxx]				Only if indicated. Safe to be administered during breastfeeding.[xxx]	✓ From 6 weeks of age.	✓ 8 weeks required between 1st and 2nd doses.[xxx]	3rd dose not required until 12 months of age.

[xxvii] Manatu Hauora (Ministry of Health) Aotearoa New Zealand. Immunisation Handbook- Influenza. Section 11.5.1- Pregnancy and breastfeeding and Section 11.5.1- Children at increased risk: Online, 2020. 11. Influenza – Immunisation Handbook 2020 | Ministry of Health NZ

[xxviii] Manatu Hauora (Ministry of Health) Aotearoa New Zealand. Immunisation Handbook- Measles (MMR). Section 12.5.4- Pregnancy and breastfeeding: Online, 2020. 12. Measles – Immunisation Handbook 2020 | Ministry of Health NZ

[xxix] Manatu Hauora (Ministry of Health) Aotearoa New Zealand. Immunisation Handbook- Rubella (MMR). Section 19.5.3- Pregnancy and breastfeeding: Online, 2020 19. Rubella – Immunisation Handbook 2020 | Ministry of Health NZ

[xxx] Manatu Hauora (Ministry of Health) Aotearoa New Zealand. Immunisation Handbook- Meningococcal disease. Section 13.5- Recommended immunisation schedule and Section 13.5.3 Pregnancy and breastfeeding. 13. Meningococcal disease – Immunisation Handbook 2020 | Ministry of Health NZ

Table 2. Vaccination recommendations- Aotearoa New Zealand

Disease/Vaccination	Planning pregnancy	Trimester 1	Trimester 2	Trimester 3	Postpartum	Infants		
						6 weeks	3 months	5 months
Monkeypox (mpox)	Not yet approved by Medsafe. May be prescribed under S29 and administered under S25 of Medicines Act 1981 for high-risk individuals (pre-exposure) and sexual contacts of people with mpox (post-exposure) vaccination. Experience in women planning pregnancy, women who are pregnant and breastfeeding is limited- **may be given after risk-benefit discussion**.[xxxi]					No advice identified specific to infants < 6 months of age.		
Pertussis (Whooping cough) dTpa combined vaccine- diphtheria, tetanus, and acellular pertussis-containing vaccine	No advice identified specific to planning pregnancy.	✓ Recommended as a single dose from 16 weeks' gestation of every pregnancy[xxxii]. Provides **protection for** . Tdap combination vaccine- diphtheria, tetanus, and pertussis.[xxxii]			✓ Vaccination is recommended for women who were **not vaccinated during pregnancy**.	✓ From 6 weeks of age. dTpa combined vaccine- diphtheria, tetanus, and acellular pertussis-containing vaccine.[xxxii]	✓ At 3 months of age.	✓ At 5 months of age.

[xxxi] Advice provided directly by the Immunisation Advisory Centre New Zealand, dated 23/01/2023, in recognition of frequent updates to eligibility criteria and availability of vaccine only under section 29 of the Medicines Act 1981. Available on request. See also- Monkeypox vaccination available to eligible people from next week | Beehive.govt.nz (current at publication)

[xxxii] Manatu Hauora (Ministry of Health) Aotearoa New Zealand. Immunisation Handbook- Pertussis (whooping cough). Section 15.5: Recommended immunisation schedule. Online, 2020. 15. Pertussis (whooping cough) – Immunisation Handbook 2020 | Ministry of Health NZ

Table 2. Vaccination recommendations- Aotearoa New Zealand

Disease/Vaccination	Planning pregnancy	Trimester 1	Trimester 2	Trimester 3	Postpartum	Infants		
						6 weeks	3 months	5 months
Pneumococcal disease	✓ Women of child-bearing age with risk condition for pneumococcal disease can receive the vaccine before a planned pregnancy.	Not routinely recommended. Only if indicated.[xxxiii]			Only if indicated. Safe to be administered during breastfeeding.[xxxiii]	✓ As early as 6 weeks of age.		✓ Second dose at **5 months of age**.[xxxiv] Booster **not required until 12 months** of age.
Rubella	Women of child-bearing age should be assessed to determine immunity to rubella.[xxxv]	✗ MMR- containing vaccines are contraindicated during pregnancy.			✓ There is no risk to mother or child in giving MMR to breastfeeding women.	Not indicated <12 months of age.		
Rotavirus	Childhood immunisation only. No concern caused by vaccine exposure during pregnancy or breastfeeding (after vaccination of the infant).[xxxvi]					✓ At 6 weeks of age.[xxxvi]	✓ At 3 months of age.	Not indicated.

[xxxiii] Manatu Hauora (Ministry of Health) Aotearoa New Zealand. Immunisation Handbook- Pneumococcal disease. Section 16.5- Recommended immunisation schedule and Section 16.5.5 Pregnancy and breastfeeding. 16. Pneumococcal disease – Immunisation Handbook 2020 | Ministry of Health NZ

[xxxiv] Manatu Hauora (Ministry of Health) Aotearoa New Zealand. Immunisation Handbook- Pneumococcal disease. Section 16.5- Recommended immunisation schedule and Section 16.5.1 Usual childhood schedule- Children aged under 5 years. 16. Pneumococcal disease – Immunisation Handbook 2020 | Ministry of Health NZ

[xxxv] Manatu Hauora (Ministry of Health) Aotearoa New Zealand. Immunisation Handbook- Rubella. Section 19.5.3- Pregnancy and breastfeeding. 19. Rubella – Immunisation Handbook 2020 | Ministry of Health NZ

[xxxvi] Manatu Hauora (Ministry of Health) Aotearoa New Zealand. Immunisation Handbook- Rotavirus. Section 18.5.1- Recommended immunisation schedule and Section 18.5.5- Pregnancy and breastfeeding: Online, 2020. 18. Rotavirus – Immunisation Handbook 2020 | Ministry of Health NZ

Table 2. Vaccination recommendations- Aotearoa New Zealand

Disease/Vaccination	Planning pregnancy	Trimester 1	Trimester 2	Trimester 3	Postpartum	Infants		
						6 weeks	3 months	5 months
Varicella (chickenpox)	✗ Avoid pregnancy for at least four weeks after varicella vaccination.[xxxvii]			✗ Contraindicated during pregnancy[xxix]	✓ Can be given safely after birth (if required) and to breastfeeding women.	Not indicated < 12 months of age.		

[xxxvii] Manatu Hauora (Ministry of Health) Aotearoa New Zealand. Immunisation Handbook- Measles (MMR). Section 12.5.4- Pregnancy and breastfeeding: Online, 2020. 12. Measles – Immunisation Handbook 2020 | Ministry of Health NZ

Table 3. Additional vaccination recommendations for higher-risk population groups- Australia and Aotearoa New Zealand

Group	Additional vaccination recommendations	Time period	Country
Aboriginal and Torres Strait Islander women and infants up to 6 months of age[xxxviii]	**BCG:** Neonates living in areas with high tuberculosis cases in NT, QLD, and northern SA (Infographic).	Within first 28 days of life.	Australia
	Japanese encephalitis: Recommended for residents of the outer islands in Torres Strait and non-residents who will be living or working on the outer islands of Torres Strait for a cumulative total of 30 days or more during December- May.	>2 months- 3 years (Table- Recommended doses of Japanese encephalitis vaccines). Infants <9 months of age should only receive the inactivated vaccine.	
	Meningococcal B: Aboriginal and Torres Strait Islander children.	At 2 months, 4 months, and 6 months of age[xxxix].	
	Pneumococcal (13vPCV): Infants living in NT, QLD, SA and WA are recommended to have an additional dose.	At 6 months of age.[xl]	Australia

[xxxviii] Australian Technical Advisory Group on Immunisation (ATAGI). Australian Immunisation Handbook- Vaccination for Aboriginal and Torres Strait Islander people, Australian Government Department of Health and Aged Care, Canberra, 2022. Vaccination for Aboriginal and Torres Strait Islander people | The Australian Immunisation Handbook (health.gov.au)

[xxxix] The Meningococcal B vaccine **is free** under the National Immunisation Program for Aboriginal and Torres Strait Islander children aged 2 months, 4 months, and 6 months (certain medical conditions). See- Meningococcal vaccine | Australian Government Department of Health and Aged Care

[xl] Australian Technical Advisory Group on Immunisation (ATAGI). Australian Immunisation Handbook- Pneumococcal disease- Recommendations- Aboriginal and Torres Strait Islander people, Australian Government Department of Health and Aged Care, Canberra, 2022. https://immunisationhandbook.health.gov.au/contents/vaccine-preventable-diseases/pneumococcal-disease#aboriginal-and-torres-strait-islander-people

Table 3. Additional vaccination recommendations for higher-risk population groups- Australia and Aotearoa New Zealand

Group	Additional vaccination recommendations	Time period	Country
Pre-term infants, born <28 weeks gestation[xli, xlii]	**Pneumococcal:** An additional dose (3 initial doses and one additional- 4 in total) of 13vPCV (13-valent pneumococcal conjugate vaccine).	Additional dose at 6 months of age for non-Indigenous infants.[xliii]	Australia
		Additional dose at 3 months of age (or as per age-appropriate catch-up schedule).[xliv]	Aotearoa New Zealand
	Pertussis: A single dose of Tdap is recommended for **primary caregivers of infants** who are patients in a Neonatal Intensive Care Unit (NICU) or special baby care unit for > 3 days and whose mother did not receive a maternal Tdap vaccination within 14 days prior to the birth (4.2.2).	Within infant's period in NICU.	Aotearoa New Zealand
	Hepatitis B: An additional dose is recommended for infants born < 32 weeks and/or with low birth weight <2000g. Preterm infants born to hepatitis B surface antigen (HBsAg) +ve mothers should receive both the vaccine and Hep B immunoglobulin.	At birth, then at 2, 4 and 6 months of age.	Australia
	Rotavirus: Premature infants can receive without correction for gestational age, including hospitalised infants who are medically stable.	From chronological age of at least 6 weeks.	Australia

[xli] Australian Technical Advisory Group on Immunisation (ATAGI). Australian Immunisation Handbook- Vaccination for preterm infants, Australian Government Department of Health and Aged Care, Canberra, 2022. Vaccination for preterm infants | The Australian Immunisation Handbook (health.gov.au)

[xlii] Manatu Hauora (Ministry of Health) Aotearoa New Zealand. Immunisation Handbook- Immunisation of special groups. Section 4.2.2 Preterm and/or low birthweight infants. 4. Immunisation of special groups – Immunisation Handbook 2020 | Ministry of Health NZ

[xliii] Australian Technical Advisory Group on Immunisation (ATAGI). Australian Immunisation Handbook- Pneumococcal disease- People with medical risk factors, Australian Government Department of Health and Aged Care, Canberra, 2022. People with medical risk factors | Pneumococcal disease | The Australian Immunisation Handbook (health.gov.au)

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Table 3. Additional vaccination recommendations for higher-risk population groups- Australia and Aotearoa New Zealand

Group	Additional vaccination recommendations	Time period	Country
Vaccination for infants born to mothers who received immunosuppressive therapy, including biological disease-modifying anti-rheumatic drugs (bDMARDs) and other biological agents[xlv, xlvi]	✕ **Any live attenuated vaccine (including rotavirus, BCG and oral polio):** Contraindicated: If an infant's mother received bDMARDs during pregnancy, especially in the 3rd trimester, it is not recommended the infant receives the rotavirus vaccine.	Seek expert advice. Decision to be made based on individual clinical data, including serum testing <15 weeks of age. If no detectable bDMARD levels, rotavirus vaccine can be safely given at this stage. If infant still has detectable bDMARD levels at 15 weeks of age, they should not receive the rotavirus vaccine. BCG vaccination is not recommended within the first few months of life. No specific interval is indicated; however, the infant (who meets neonatal BCG eligibility criteria[xlvii]) should be at least 6 months of age before receiving a BCG vaccine.	Australia
		Assessed on a risk-benefit, case by case (Table 4.2) basis with specialist advice required (18.6.2). If an infant turns 15 weeks of age before Dose 1 of the rotavirus vaccine, they will not be able to receive any rotavirus vaccine doses.	Aotearoa New Zealand

[xlv] Australian Technical Advisory Group on Immunisation (ATAGI). Australian Immunisation Handbook- Vaccination for women who are planning pregnancy, pregnant or breastfeeding- live attenuated vaccines and immunosuppressive therapy. Australian Government Department of Health and Aged Care, Canberra, 2022. https://immunisationhandbook.health.gov.au/contents/vaccination-for-special-risk-groups/vaccination-for-women-who-are-planning-pregnancy-pregnant-or-breastfeeding#inactivated-vaccines-and-immunosuppressive-therapy

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